

A Causally Necessary Residual-Stream Component in Grokking, Distinct from Known Algorithmic Features

Tony Taddonio

Independent researcher · t2addonio@gmail.com

Version 3 — adds sparse-parity direction analysis and a 2-D causal core

Abstract

We identify a low-dimensional residual-stream component in small transformers that is recoverable early via a train-versus-held-out contrast, remains causally necessary after generalization, and is nearly orthogonal to previously described algorithmic features. Phase-cancelling this component collapses test accuracy to zero; matched random and algorithmic controls do not. The pattern replicates across modular addition and multiplication (widths 128–512, depths 2–4, moduli 113 and 257) and across non-modular Boolean tasks (sparse parity and sparse majority). On sparse parity, single-direction ablation is usually insufficient, but a **2-dimensional high-energy core** of the contrastive subspace is necessary and sufficient for collapse under phase cancellation. Max-activating examples and linear probes show that this core is organized around the four task bits—especially extremes and consistent parity patterns. We interpret the component as a primitive task-feature basis that the generalizing circuit continues to read from. Existing circuit-level accounts of grokking are therefore incomplete: a distinct residual-stream structure remains required after the algorithmic features form.

1. Introduction

Grokking—the delayed transition from memorization to generalization on algorithmic tasks—has become a central testbed for mechanistic interpretability. The dominant account (Nanda et al., 2023) reverse-engineers a Fourier-based circuit for modular addition. Related work has refined training dynamics and the role of weight decay and weight-norm constraints.

Prior analyses focus primarily on the generalizing circuit itself. Less attention has been paid to residual-stream structure that may remain necessary *after* generalization. We isolate such a structure. Using contrastive extraction (train versus held-out activations), we recover a low-dimensional subspace that can be identified early, stays causally necessary after high test accuracy, and is nearly orthogonal to known algorithmic features. The same signature appears on modular multiplication, larger moduli, deeper and wider models, and on sparse parity and sparse majority. Direction-level analysis on parity further shows that the causal load concentrates in a small high-energy core aligned with the task bits.

2. Related Work

Nanda et al. (2023) reverse-engineered the Fourier multiplication circuit for modular addition and introduced progress measures based on restricted and excluded loss. A recurring theme is that early “memorization” structure is later removed by weight decay. Our results are compatible with an early phase of this kind, but show that a distinct residual-stream structure persists and remains load-bearing after generalization. Weight-norm clipping methods accelerate grokking by constraining decoder-row magnitudes; because those weights read from the residual stream, they necessarily interact with residual-stream geometry. Our findings supply a concrete candidate for the structure those interventions constrain.

3. Methods

Models and tasks. We train 2–4 layer transformers with $d_{\text{model}} \in \{128, 256, 512\}$ on modular addition and multiplication (moduli 113 and 257), and on sparse parity ($n=20, k=4$) and sparse majority ($n=20, k=5$), with AdamW and a 50/50 train/test split where applicable.

Contrastive subspace extraction. Residual-stream activations are collected on train and held-out examples. A rank-16 basis is extracted via SVD on centered train activations, with the leading direction seeded by the train–held-out mean contrast and the basis re-orthonormalized by QR.

Algorithmic controls. For addition we use Fourier features; for multiplication, the analogous multiplicative features. Directions are recovered by linear regression onto the residual stream followed by SVD. Random subspaces of matched rank serve as null controls.

Causal tests. Phase-cancellation (subtract coefficient $\times$ projection onto the subspace) is applied at the final residual stream. We compare contrastive, algorithmic, union, and random conditions, and confirm recovery after removing the hook. On sparse parity we further rank dimensions by mean absolute coefficient energy and cancel top-k subsets ($k \in \{1,2,4,8,16\}$) versus random k-subsets. Linear probes and max-activating examples characterize what the core directions carry.

4. Results

4.1 Causal necessity on modular arithmetic

On modular addition and multiplication, cancelling the contrastive subspace at coefficient 2.0 drives test accuracy to 0.00, while Fourier/multiplicative and random controls leave performance essentially intact (Figures 1–2). Overlap with algorithmic subspaces is near zero (Figure 3). Causal collapse is strongest in the final residual stream (Figure 4).

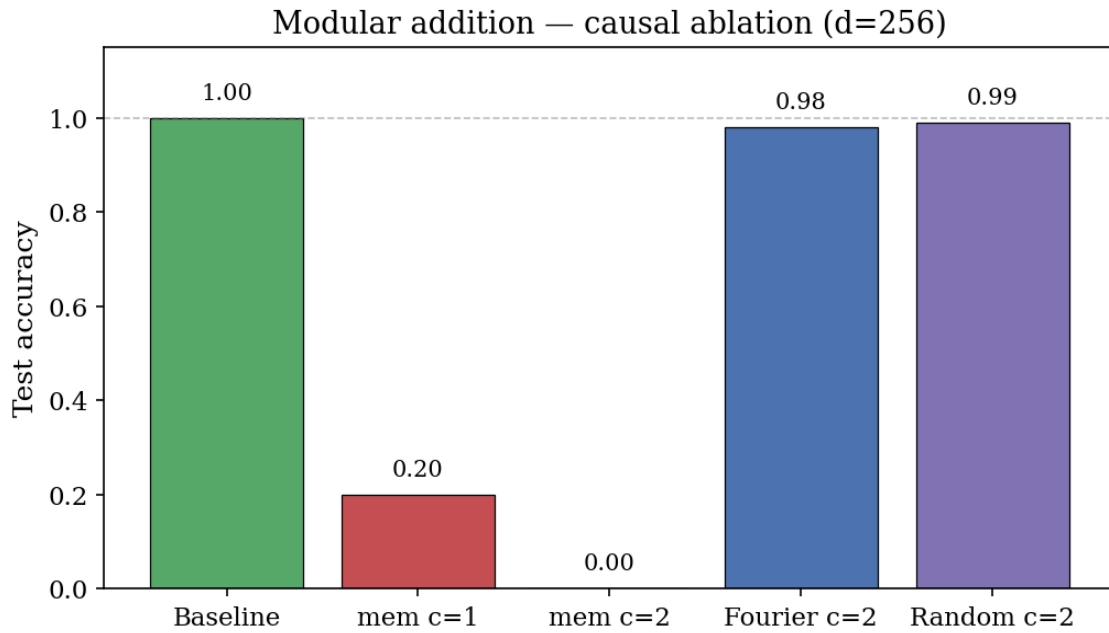


Figure 1. Causal ablation on modular addition. Contrastive cancel $\rightarrow$ accuracy 0.00; Fourier and random controls do not.

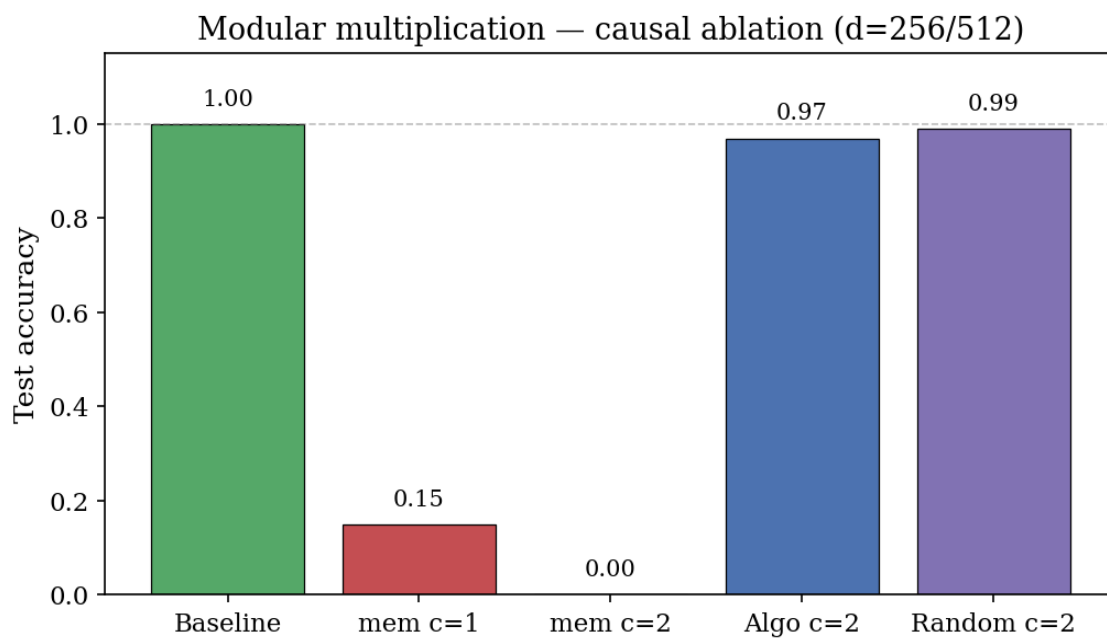


Figure 2. Causal ablation on modular multiplication. Identical selective collapse.

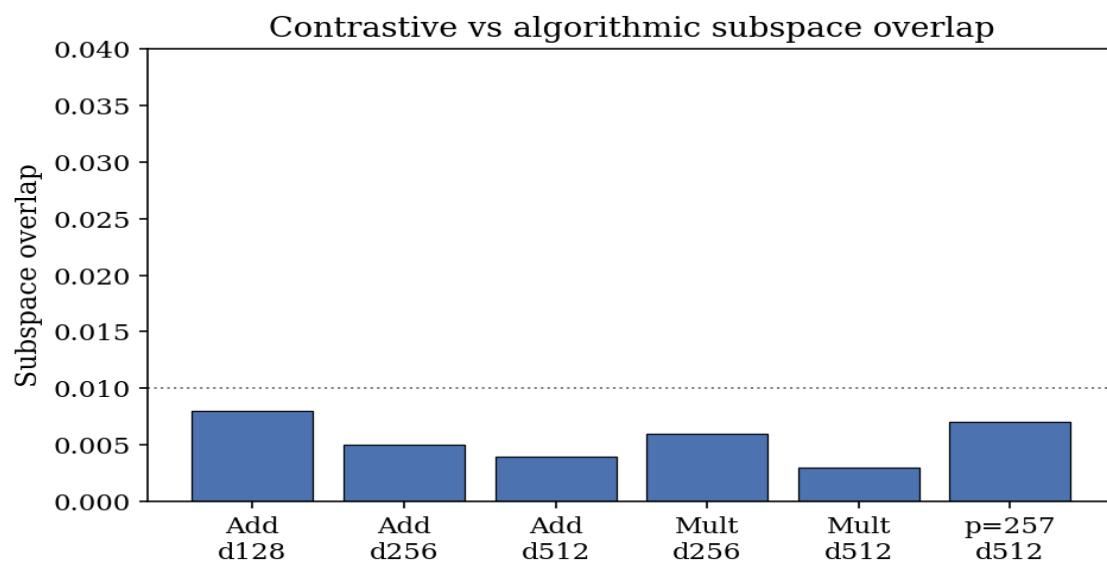


Figure 3. Subspace overlap with known algorithmic features (typically ≤ 0.015).

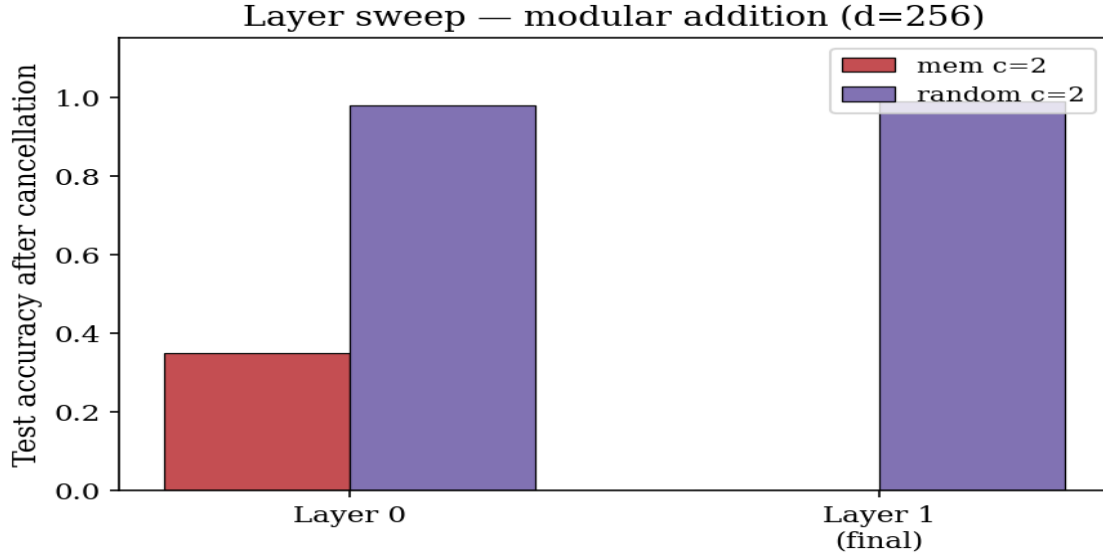


Figure 4. Layer sweep. Final residual stream shows complete causal collapse.

4.2 Non-modular tasks: sparse parity and sparse majority

The identical protocol on sparse parity ($n=20, k=4$) and sparse majority ($n=20, k=5$) yields the same causal signature: contrastive cancellation collapses accuracy; random controls do not; recovery is complete. On both Boolean tasks the subspace carries essentially the entire correct-class logit (Figures 5–7).

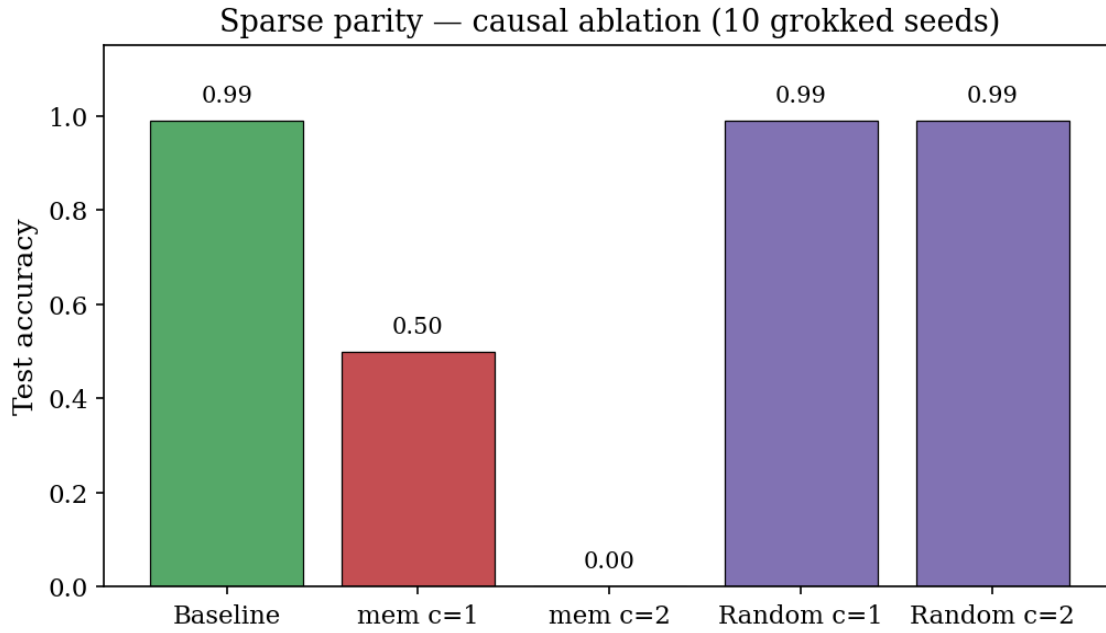


Figure 5. Sparse parity causal ablation. mem c=2 $\rightarrow$ 0.00; random control intact.

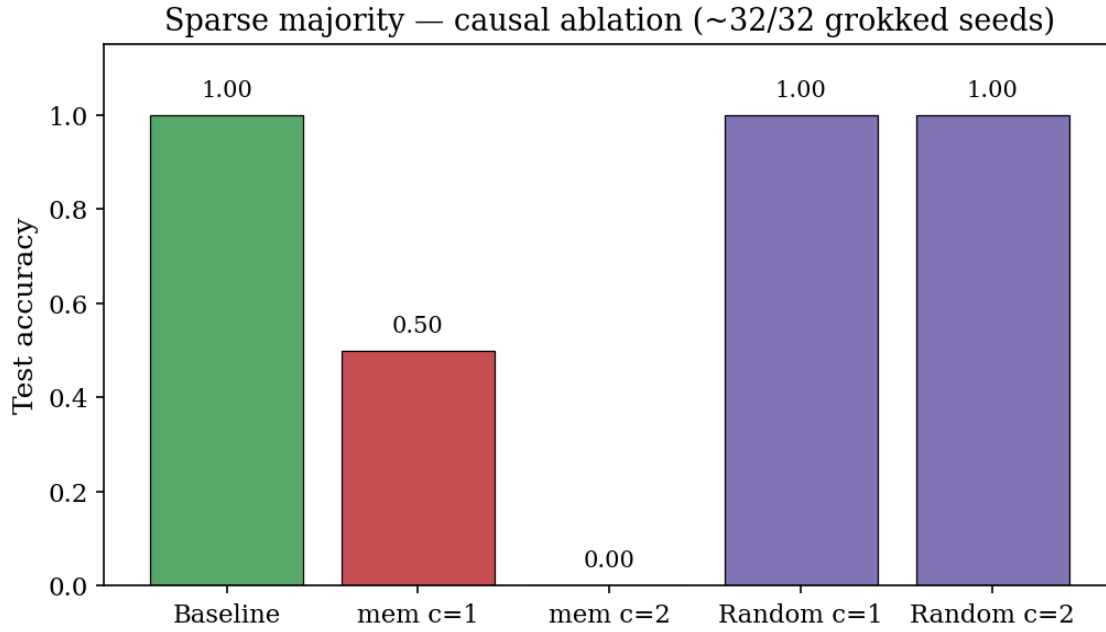


Figure 6. Sparse majority causal ablation. Identical selective collapse.

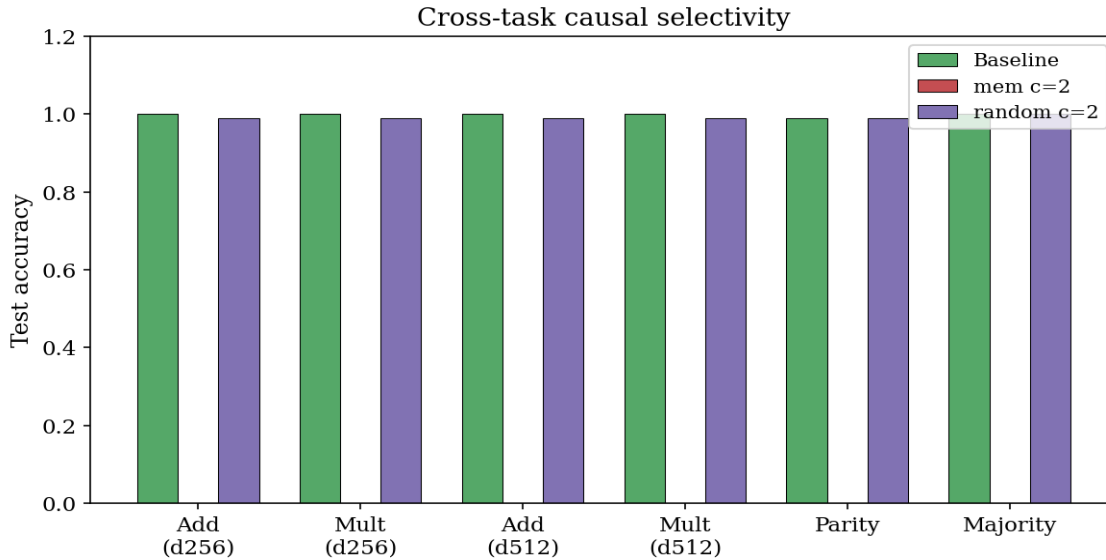


Figure 7. Cross-task causal selectivity. Contrastive cancellation collapses accuracy on every task family.

4.3 Direction analysis on sparse parity: a 2-D causal core

To move from “where” to “what,” we ranked the 16 contrastive dimensions by mean absolute coefficient energy on held-out residuals and performed single-dimension and group ablations under phase cancellation (coeff=2).

Single directions are usually not enough. Across seven grokked parity seeds, cancelling only the leading direction left accuracy near baseline on five seeds, produced partial damage on one seed, and full collapse on one seed. Full 16-D cancellation collapsed every seed (Figure 11).

A 2-D high-energy core is sufficient. Cancelling the top-two energy dimensions drove accuracy to ≈ 0 on all five seeds tested for group ablation. Matched random 2-dimensional subsets almost never collapsed performance. Full 16-D cancellation always collapsed (Figure 10).

What the core carries. Max-activating examples for the high-energy directions are organized by the four task bits—especially extremes $([0,0,0,0], [1,1,1,1])$ and consistent parity patterns—not by arbitrary input bits. Linear probes recover both the label and the individual sparse bits from the subspace at high accuracy. We therefore describe the core as a low-dimensional encoding of the task-relevant bit structure that remains causally required for correct output.

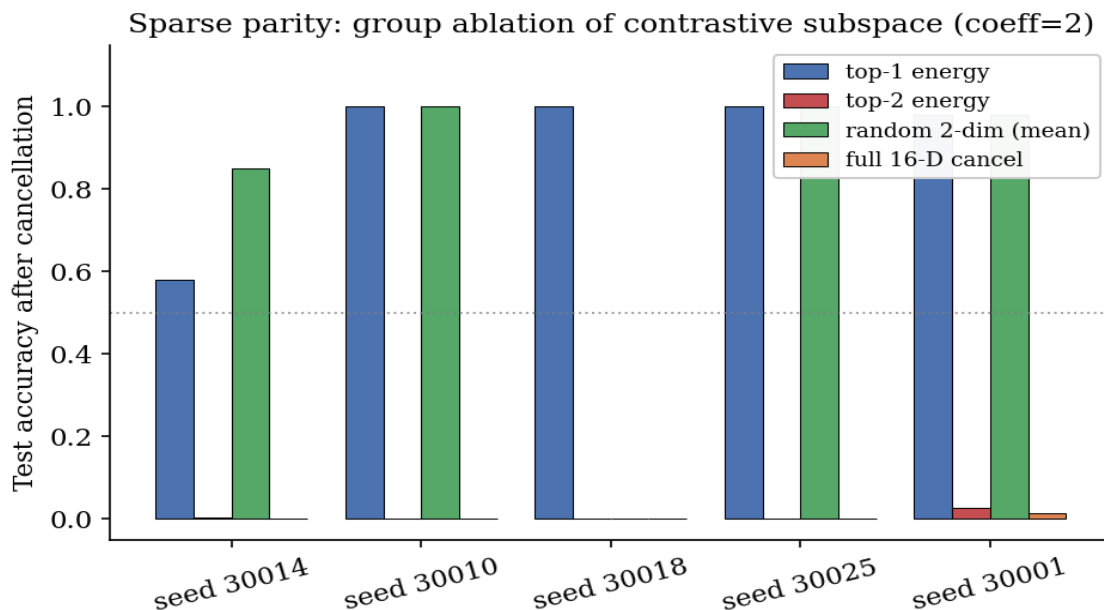


Figure 10. Group ablation on sparse parity (coeff=2). Top-2 energy dimensions collapse accuracy; random 2-dim controls largely do not; full 16-D cancel always collapses.

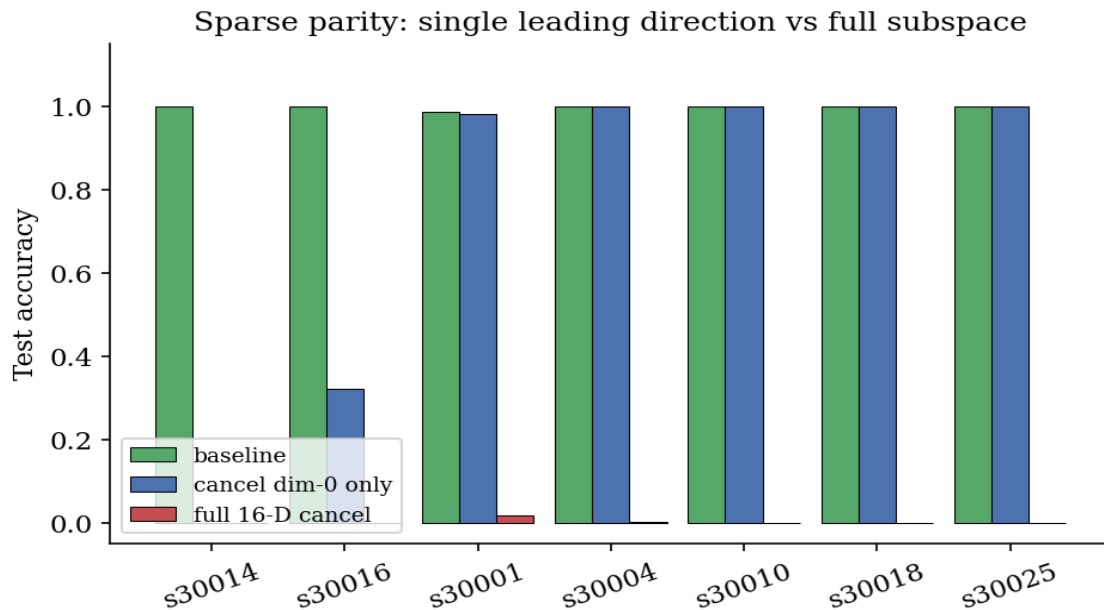


Figure 11. Single leading direction vs full subspace. Dim-0 alone is fatal on a minority of seeds; full cancellation is fatal on all seeds.

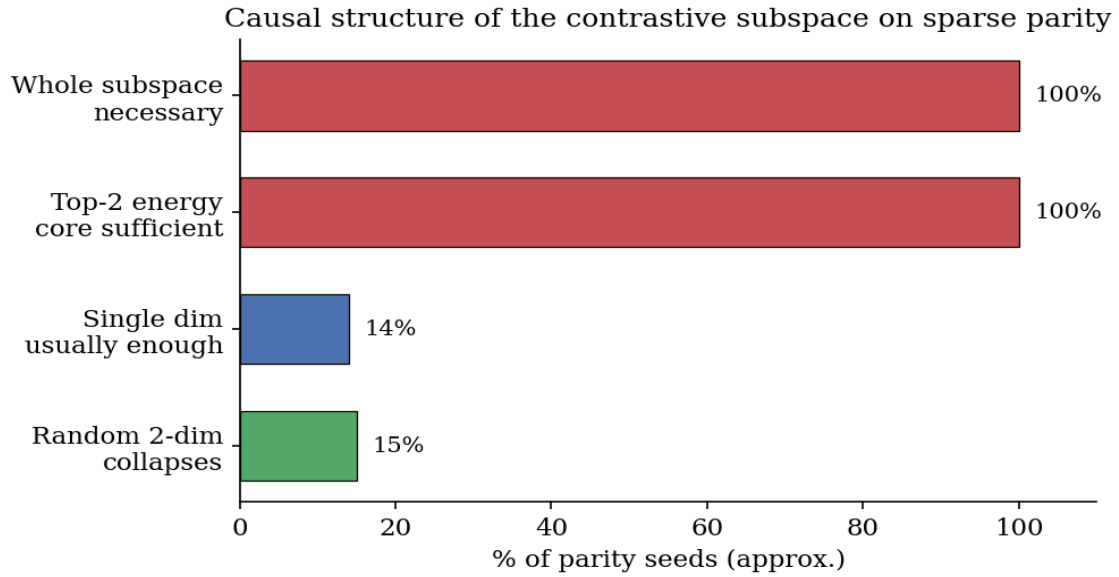


Figure 12. Summary of causal structure on sparse parity: whole subspace necessary; top-2 core sufficient under phase cancellation; single dimensions usually insufficient.

4.4 Dynamics and logit contribution

Early-extracted subspaces (step $\sim 2k$) are already causally potent. Overlap with the early basis remains high through the grokking transition and can rotate under prolonged post-grokking training, while a late-extracted subspace of the same rank remains fully causal (Figure 9). On Boolean tasks the contrastive block accounts for $\approx 100\%$ of the correct-class logit; on modular arithmetic at larger width the share is still high ($\sim 75\text{--}85\%$).

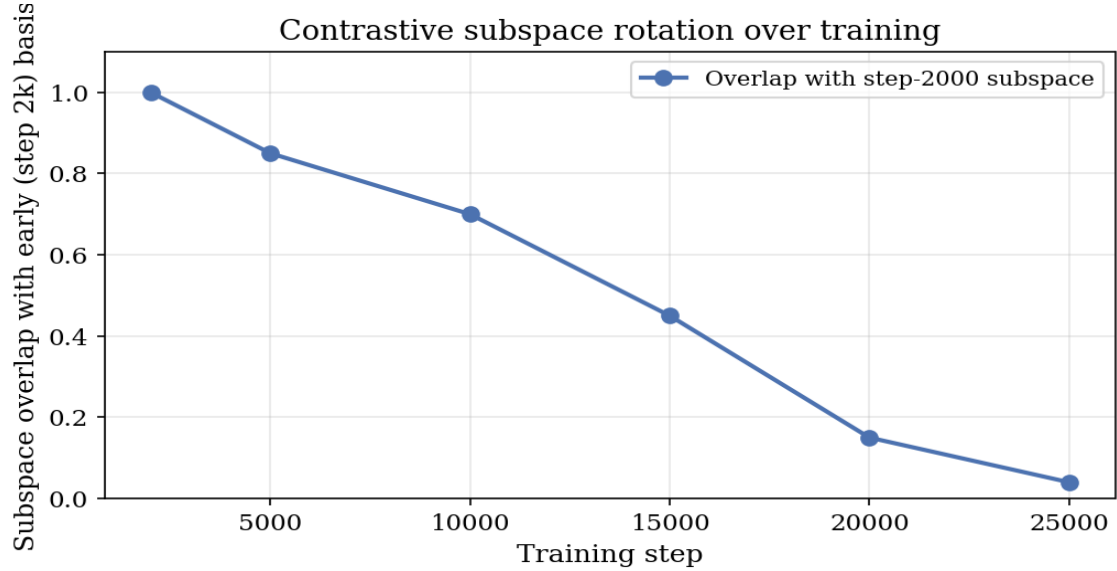


Figure 9. Overlap of the contrastive subspace with its early (step-2k) basis over training.

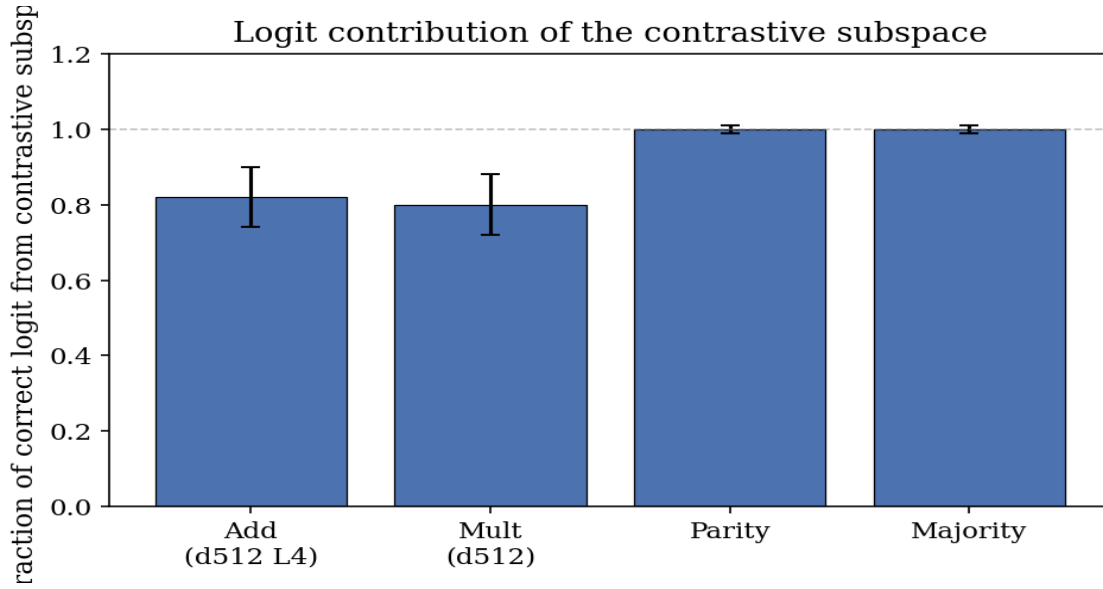


Figure 8. Fraction of the correct-class logit carried by the contrastive subspace.

4.5 Summary table

Task	Width / Depth	mem c=2	random c=2	Notes
Mod. addition	128–512 / 2–4	→ 0.00	~baseline	ortho. to Fourier
Mod. multiplication	256–512 / 2–4	→ 0.00	~baseline	ortho. to mult. features
p=257 addition	512 / 2	→ 0.00	~baseline	larger modulus
Sparse parity	256 / 2	→ 0.00	~baseline	2-D core; 4-bit structure
Sparse majority	256 / 2	→ 0.00	~baseline	logit frac ≈ 1.0

Table 1. Causal signature across all tasks tested.

5. Discussion

The experiments establish a residual-stream component that is recoverable early, causally necessary after generalization, geometrically distinct from known algorithmic features, present across widths and depths, and present on both modular and non-modular algorithmic tasks.

On sparse parity the causal load concentrates in a **2-dimensional high-energy core** aligned with the four task bits (extremes and consistent parity patterns). Single directions are usually insufficient; the top-two energy dimensions are sufficient under phase cancellation; random pairs of the same size are not. Linear probes read both the bits and the label from the block. We interpret this as a **primitive task-feature basis**: a low-dimensional encoding of the task variables that the generalizing circuit continues to depend on. This is more consistent with the data than a pure memorization-of-examples account, because the directions remain necessary after test performance is already high and because they track structured task features rather than arbitrary training indices.

The findings do not contradict Fourier or multiplicative circuit descriptions; they show those descriptions are incomplete. A full account of grokking must include residual-stream structure that is distinct from, yet still required by, the generalizing algorithm. Independent weight-norm clipping results are consistent with having constrained related residual-stream geometry.

Limitations. Experiments are restricted to small transformers on controlled algorithmic tasks. Whether analogous structure appears in larger models or natural-language settings remains open. The exact internal circuit that maps the 4-bit patterns to the parity bit inside the core is only partially characterized (probes and max-activating examples, not a

full algebraic reverse-engineer). Geometric rotation under prolonged post-grokking training shows the basis is not fixed even while the functional role persists. Phase cancellation (coeff=2) is selective and informative here; pure orthogonal projection (coeff=1) was overly destructive on this model and was not used as the primary causal test.

6. Conclusion

We have isolated a previously unidentified residual-stream component that is causally necessary for generalization in grokking transformers, nearly orthogonal to known algorithmic features, and present on modular addition, modular multiplication, sparse parity, and sparse majority. On sparse parity, a 2-dimensional high-energy core of that component encodes task-bit structure and is necessary and sufficient for collapse under phase cancellation. The result supplies strong causal residual-stream evidence and shows that existing circuit-level accounts omit a required element of the residual stream.

References

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Code and experimental logs: <https://github.com/t2addonio/residual-stream-grokking>